Hemophilia B is a blood clotting disorder caused by a factor IX (FIX) deficiency. FIX is a 57-kDa, vitamin K-dependent protease that activates factor X, leading to the conversion of prothrombin to thrombin for propagation of the clotting cascade. FIX catalyzes the interconversion of three forms of vitamin K, as shown in Figure 1.

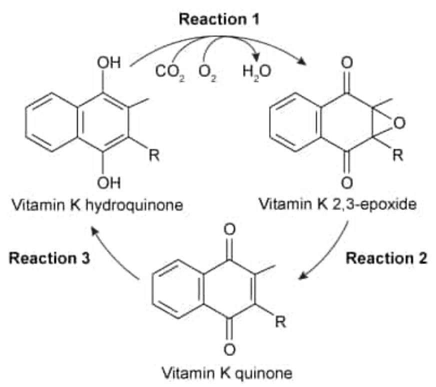


Figure 1 Oxidation of vitamin K

During Reaction 1, the side chain of an amino acid residue in FIX is carboxylated, resulting in the structure shown in Figure 2.

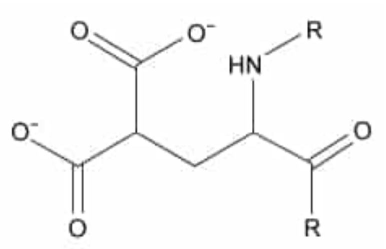


Figure 2 Structure of a carboxylated amino acid residue in FIX

To further analyze the γ-carboxyglutamic acid-rich domain of FIX, an analogous synthetic peptide composed of matching residues 1 through 49 on FIX was evaluated by proton nuclear magnetic resonance (^1H NMR) spectroscopy. Analysis of the proton chemical shift before the addition of metal ions suggested that the synthetic peptide contained normal structural elements. When calcium ions were added, the height of a peak at approximately 1 ppm was significantly diminished and a new peak appeared farther downfield.

The synthetic peptide analog was then placed in solution with vitamin K hydroquinone and cofactors required for vitamin K oxidation. The three forms of vitamin K were collected and evaluated under high-performance liquid chromatography (HPLC) using hexane as the mobile phase. The results are shown in Figure 3.

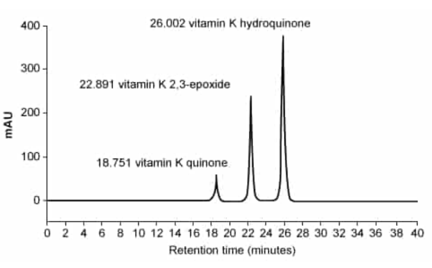


Figure 3 Results of HPLC separation

Based on the NMR results described in the passage, what effect does the addition of calcium ions have on the protons that corresponded to the 1 ppm peak?

◯ A. Calcium ions cause additional splitting of the 1 ppm peak. [7%]

◯ B. Calcium ions reduce the splitting of the 1 ppm peak. [3%]

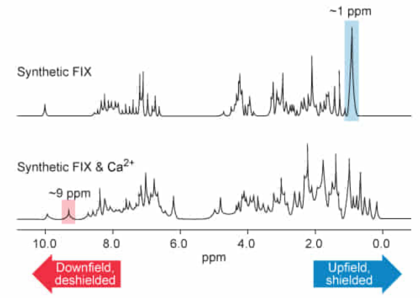
◯ C. Calcium ions result in shielding of the protons that corresponded to the 1 ppm peak. [21%]

✔ ◉ D. Calcium ions result in deshielding of the protons that corresponded to the 1 ppm peak. [67%]

Correct
68% answered correctly

Explanation:

Effect of adding calcium ions to FIX on a ^{1}H NMR signal



Addition of Ca^{2+} to FIX **deshields** the protons that correspond to the peak at 1 ppm, causing the signal to move downfield.

Proton nuclear magnetic resonance spectroscopy (^{1}H NMR) detects hydrogen atoms (protons) in a molecule by applying an external magnetic field and radio frequencies to a sample. NMR measures the relative resonance frequencies of protons, which are displayed as chemical shifts on an NMR spectrum. The chemical shift of a proton depends on its magnetic environment. Protons surrounded by high electron density are said to be shielded from magnetic effects and have small chemical shifts (ie, they are upfield on the spectrum), whereas protons with low electron density are deshielded and have large chemical shifts (ie, they are downfield).

In general, the magnitude of shielding is dependent on the local chemical environment. For instance, protons within or near methyl groups are highly **shielded** and have a **smaller (upfield) chemical shift**. Nearby electronegative atoms, such as oxygen and nitrogen, **pull electron density away** from **(deshield)** the proton and have a downfield chemical shift. For example, double bonds can cause an anisotropic effect, which deshields protons, resulting in a downfield chemical shift relative to alkyl groups.

The passage states that a ^{1}H NMR peak at approximately 1 ppm is diminished when calcium ions are added, with a new peak appearing farther downfield. The 1 ppm peak has a comparatively small chemical shift (ie, it is found upfield), which suggests that protons corresponding to this peak are more shielded by the surrounding electron density. The reduction of this peak and the **appearance of a downfield peak** upon addition of calcium ions suggests that **calcium ions cause deshielding** of the corresponding protons.

(Choices A and B) Peak splitting is related to the number of hydrogen atoms covalently bonded to carbons that are *adjacent to* the C-H bonds of interest. Calcium ions are unlikely to change the covalent bonds in FIX, and the passage does not mention any changes in splitting.

(Choice C) The passage states that the new peak is downfield of the original peak. If calcium ions caused shielding, the new peak would have appeared *upfield* of 1 ppm.

Educational objective:
Proton nuclear magnetic resonance detects protons in a molecule by applying an external magnetic field and radio frequencies to a sample. Protons shielded by nearby electrons have smaller chemical shifts (upfield). Deshielded protons, such as those near electronegative atoms, have larger chemical shifts (downfield).

Time spent: 0 sec
Subject: Organic Chemistry

QID: 400291
System: Introduction to Organic Chemistry

Hemophilia B is a blood clotting disorder caused by a factor IX (FIX) deficiency. FIX is a 57-kDa, vitamin K-dependent protease that activates factor X, leading to the conversion of prothrombin to thrombin for propagation of the clotting cascade. FIX catalyzes the interconversion of three forms of vitamin K, as shown in Figure 1.

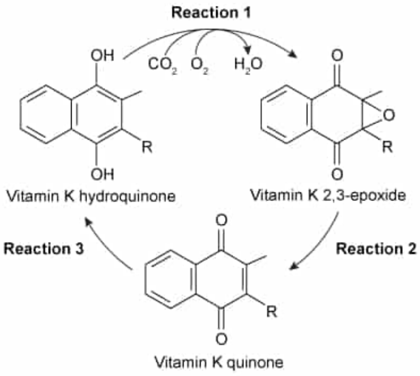


Figure 1 Oxidation of vitamin K

During Reaction 1, the side chain of an amino acid residue in FIX is carboxylated, resulting in the structure shown in Figure 2.

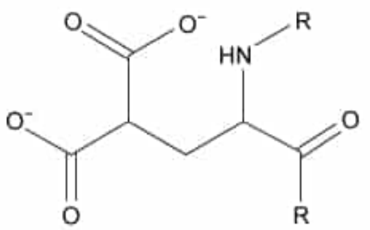


Figure 2 Structure of a carboxylated amino acid residue in FIX

To further analyze the γ-carboxyglutamic acid-rich domain of FIX, an analogous synthetic peptide composed of matching residues 1 through 49 on FIX was evaluated by proton nuclear magnetic resonance (^1H NMR) spectroscopy. Analysis of the proton chemical shift before the addition of metal ions suggested that the synthetic peptide contained normal structural elements. When calcium ions were added, the height of a peak at approximately 1 ppm was significantly diminished and a new peak appeared farther downfield.

The synthetic peptide analog was then placed in solution with vitamin K hydroquinone and cofactors required for vitamin K oxidation. The three forms of vitamin K were collected and evaluated under high-performance liquid chromatography (HPLC) using hexane as the mobile phase. The results are shown in Figure 3.

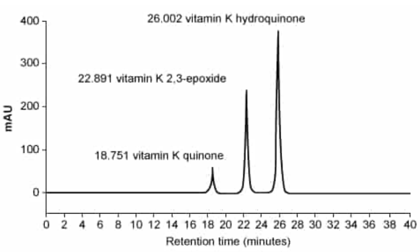


Figure 3 Results of HPLC separation

Suppose the HPLC experiment used to generate Figure 3 is repeated, but the sample is contaminated. If the results include a new peak with a retention time of 23.421 minutes, the contaminant most likely:

- ◯ A. has a lower affinity for the stationary phase than vitamin K quinone. [7%]
- ◯ B. has a lower affinity for the stationary phase than vitamin K 2,3-epoxide. [9%]

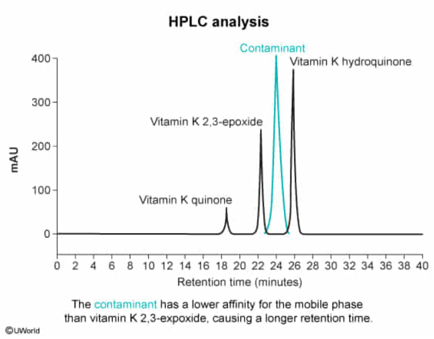
微信Mai_ange

✔ C. has a lower affinity for the mobile phase than vitamin K 2,3-epoxide. [58%]

D. has a lower affinity for the mobile phase than vitamin K hydroquinone. [24%]

Correct
59% answered correctly

Explanation:



High-performance liquid chromatography **(HPLC)** is a column-based analytical technique used to **separate, quantify, and characterize** components of a mixture. Like other forms of chromatography, HPLC's mobile phase (eg, hexane) migrates through a stationary phase (eg, silica) and carries the compounds of interest with it.

Retention indicates the length of time for an isolated compound to traverse the HPLC column and is correlated to affinity for the mobile phase versus the stationary phase (ie, **longer retention time** indicates **greater affinity** for the **stationary phase** and **less affinity** for the **mobile phase**). Once components are separated, a detector measures the amount of light entering and passing through each isolated compound, providing the absorption that is represented by a peak and measured in milli-absorption units (mAUs).

In the given scenario, the new peak (which corresponds to the contaminant) has a longer retention time than vitamin K 2,3-epoxide, indicating that the **contaminant** has a **lower affinity** for the **mobile phase** (and a higher affinity for the stationary phase) **than vitamin K 2,3-epoxide** has.

(Choice A) The contaminant has a longer retention time than vitamin K quinone. Because the contaminant was retained on the column longer, it must have a *greater* affinity for the stationary phase than vitamin K quinone has.

(Choice B) Although the contaminant has a lower affinity for the *mobile* phase than vitamin K 2,3-epoxide has, the opposite is true of their relative affinities for the *stationary* phase.

(Choice D) The contaminant has a shorter retention time than vitamin K hydroquinone, meaning it has a *greater* affinity for the mobile phase than vitamin K hydroquinone has.

Educational objective:
High-performance liquid chromatography is a column-based analytical technique used to separate, quantify, and identify components of a mixture. Compounds are separated based on their affinity for the mobile or stationary phase. Compounds with longer retention times have a greater affinity for the stationary phase and a lower affinity for the mobile phase compared to compounds with shorter retention times.

Time spent:0 sec
Subject:
Organic Chemistry

QID:400292
System:
Introduction to Organic Chemistry

Hemophilia B is a blood clotting disorder caused by a factor IX (FIX) deficiency. FIX is a 57-kDa, vitamin K-dependent protease that activates factor X, leading to the conversion of prothrombin to thrombin for propagation of the clotting cascade. FIX catalyzes the interconversion of three forms of vitamin K, as shown in Figure 1.

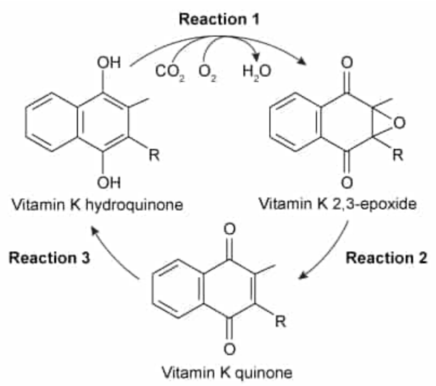


Figure 1 Oxidation of vitamin K

During Reaction 1, the side chain of an amino acid residue in FIX is carboxylated, resulting in the structure shown in Figure 2.

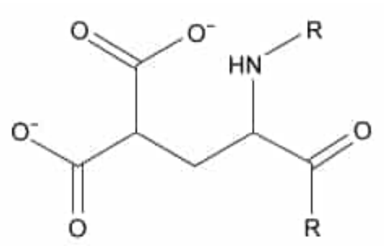


Figure 2 Structure of a carboxylated amino acid residue in FIX

To further analyze the γ-carboxyglutamic acid-rich domain of FIX, an analogous synthetic peptide composed of matching residues 1 through 49 on FIX was evaluated by proton nuclear magnetic resonance (^{1}H NMR) spectroscopy. Analysis of the proton chemical shift before the addition of metal ions suggested that the synthetic peptide contained normal structural elements. When calcium ions were added, the height of a peak at approximately 1 ppm was significantly diminished and a new peak appeared farther downfield.

The synthetic peptide analog was then placed in solution with vitamin K hydroquinone and cofactors required for vitamin K oxidation. The three forms of vitamin K were collected and evaluated under high-performance liquid chromatography (HPLC) using hexane as the mobile phase. The results are shown in Figure 3.

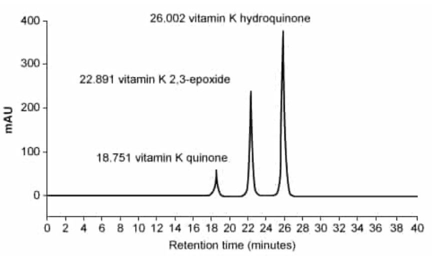


Figure 3 Results of HPLC separation

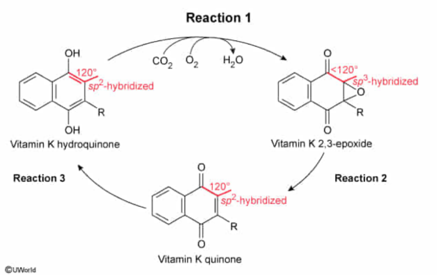
Many of the bonds in vitamin K have angles of 120°. Which reaction in Figure 1 results in one or more new bonds with angles of less than 120°?

A. Reaction 1 only [77%]

◯ B. Reaction 2 only [9%]
◯ C. Reaction 3 only [4%]
◯ D. Reactions 1 and 2 only [8%]

Correct
77% answered correctly

Explanation:



Valence shell electron pair repulsion (VSEPR) theorizes that atoms within a molecule tend to achieve a **geometry that minimizes repulsion** between electrons by maximizing distance between lone pairs and bonds. As a result, only a limited number of expected molecular shapes exists. These shapes are defined by an atom's hybrid orbitals, which depend on the number and position of lone electron pairs and sigma bonds.

In general, ***sp*^{3}-, *sp*^{2}-, and *sp*-hybridized atoms** have **bond angles of 109.5°, 120°, and 180°**, respectively. Lone electron pairs or ring strain may distort predicted bond angles. For example, **water** displays a **bond angle of 104.5°** due to **increased electron repulsion** of its **lone pairs** despite its *sp*^{3}-hybridized orbitals, which would normally suggest a bond angle of 109.5°.

All the carbon atoms in the ring structures of vitamin K have bond angles of 120° except the carbons involved in the epoxide. The epoxide carbons are *sp*^{3}-hybridized, which normally predicts bond angles of 109.5°. These angles are slightly distorted by the ring strain induced by the epoxide but are still less than 120°. Because the epoxide is formed by Reaction 1, **only Reaction 1** is expected to cause **bond angles** to **change from 120° to less than 120°**.

(Choices B and D) Reaction 2 converts the epoxide carbons back into *sp*^{2}-hybridized carbons. This process *restores* the bond angles to 120°.

(Choice C) All carbons in the rings of both the reactant and the product in Reaction 3 have bond angles of 120°.

Educational objective:
Valence shell electron pair repulsion theorizes that atoms within a molecule strive to achieve a geometry that minimizes electron repulsion. Hybridization of *sp*, *sp*^{2}, and *sp*^{3} correlates with 180°, 120°, and 109.5° bond angles, respectively. Factors such as lone electron pairs and ring strain can distort these expected angles.

Time spent:0 sec	**QID**:400293
Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry

Hemophilia B is a blood clotting disorder caused by a factor IX (FIX) deficiency. FIX is a 57-kDa, vitamin K-dependent protease that activates factor X, leading to the conversion of prothrombin to thrombin for propagation of the clotting cascade. FIX catalyzes the interconversion of three forms of vitamin K, as shown in Figure 1.

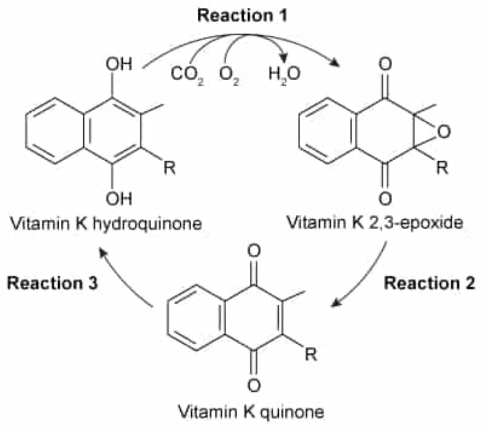


Figure 1 Oxidation of vitamin K

During Reaction 1, the side chain of an amino acid residue in FIX is carboxylated, resulting in the structure shown in Figure 2.

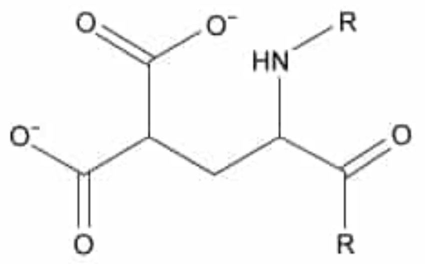


Figure 2 Structure of a carboxylated amino acid residue in FIX

To further analyze the γ-carboxyglutamic acid-rich domain of FIX, an analogous synthetic peptide composed of matching residues 1 through 49 on FIX was evaluated by proton nuclear magnetic resonance (^{1}H NMR) spectroscopy. Analysis of the proton chemical shift before the addition of metal ions suggested that the synthetic peptide contained normal structural elements. When calcium ions were added, the height of a peak at approximately 1 ppm was significantly diminished and a new peak appeared farther downfield.

The synthetic peptide analog was then placed in solution with vitamin K hydroquinone and cofactors required for vitamin K oxidation. The three forms of vitamin K were collected and evaluated under high-performance liquid chromatography (HPLC) using hexane as the mobile phase. The results are shown in Figure 3.

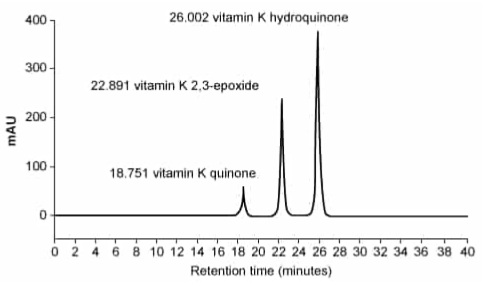


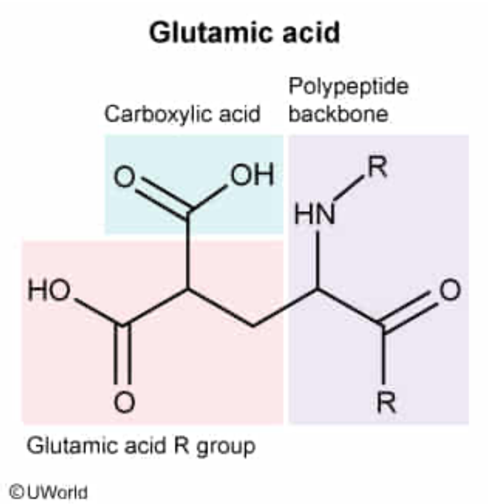
Figure 3 Results of HPLC separation

Based on Figure 2, which amino acid residue in FIX most likely undergoes carboxylation during Reaction 1?

- A. Aspartate [17%]
- B. Asparagine [7%]
- C. Glutamine [9%]
- ✔ D. Glutamate [65%]

Correct
64% answered correctly

Explanation:



The carboxylation (addition of a carboxyl group) in Reaction 1 forms γ-carboxyglutamate, shown in Figure 2. Formation of this uncommon amino acid leads to the activation of FIX. **Glutamate (Glu, E)** is a **negatively charged, polar** amino acid with a three-carbon side chain containing one **carboxyl group**. Figure 2 shows an amino acid side chain that contains three carbons in the main chain and *two* carboxyl groups, one of which is at a branch point. This residue could most easily be produced by adding a carboxyl group to the side chain of glutamate.

(Choice A) Aspartate (Asp, D) is a negatively charged, polar amino acid with a *two*-carbon side chain containing a carboxyl group.
(Choices B and C) Asparagine (Asn, N) and glutamine (Gln, Q) are uncharged, polar amino acids that contain an amide (not a carboxyl group) on their side chain. Asparagine has a two-carbon side chain, and glutamine has a three-carbon side chain.

Educational objective:
Glutamate is a negatively charged, polar amino acid with a three-carbon side chain containing a carboxylic acid.

Time spent:0 sec
Subject:
Organic Chemistry

QID:400294
System:
Introduction to Organic Chemistry